Learning Whom to Trust from Feedback: Raw, Summarized and Described Advisor Accuracy in Vision-Language Models

Anonymous submission

Abstract

[DRAFT: development episodes (16) and four models; replace with held-out results] Adapting to a volatile environment, as people do when they learn whom to trust, is a hallmark of context-dependent learning. We ask whether vision-language models (VLMs), used without fine-tuning, can learn an advisor's accuracy by in-context learning from trial-by-trial feedback, and which step fails if they cannot. We adapt a human advice-taking task with stable and volatile advisor accuracy and give the same history in formats that remove its processing steps one at a time: judging who was correct on each trial, integrating these scores, and using the estimate. Given the raw record of reports and outcomes, models chose at or below chance and their choices tracked the previous outcome rather than the advisors. Stating who was correct on each trial raised accuracy by 15–17 percentage points, to the level of a summary of counts, and described accuracy was used almost perfectly. A Bayesian learner that describes human participants in the original task was more accurate on the same episodes than any model in any format. The bottleneck is judging raw outcomes, not integrating them. **[TODO: finalize with held-out results and the human comparison]**

Introduction

People learn whom to trust from feedback. When two advisors disagree and the outcome is revealed afterwards, people keep track of how often each advisor was right, and they update this estimate faster when the advisor's accuracy is volatile (Behrens et al. 2008); advisor reputations form quickly from a track record (Yaniv and Kleinberger 2000; Bonaccio and Dalal 2006). Because this advice-taking task comes from human neuroscience, its behavioral signatures can be measured in a model in the same way and compared with people.

Agents built on large language models (LLMs) and vision-language models (VLMs) face the same problem when they combine tool outputs, retrieved documents or messages from other agents whose reliability can change. Prior work asked whether LLMs weight sources by reliability when it is fixed (Ashkinaze et al. 2026), flips once (Zhou et al. 2026) or breaks after scripted failures (Chen 2026), and proposed memories that track reliability outside the model (Feng, Yang, and Poria 2026). Whether a model without fine-tuning can learn a changing, non-stationary accuracy by in-context learning from trial-by-trial feedback, and which part of the computation fails if it cannot, is open. This is context-dependent learning without plasticity: a model whose weights are not updated can adapt only through what its context contains.

The format of past evidence is known to matter. LLMs choose differently when outcome probabilities are described than when they are experienced (Ge, Zhang, and Vorobeychik 2026), explore well in bandits only with summarized histories (Krishnamurthy et al. 2024), and integrate evidence with strong recency (Xue, Wilson, and Xiong 2026). These studies do not separate which processing step fails, do not involve advisors, and are text-only.

We decompose learning whom to trust into three steps (Fig. 1b): (i) judging which advisor was correct on each trial, (ii) integrating these scores across trials, and (iii) using the estimate. Five VLMs see the same history in formats that remove these steps one at a time, and a description condition states each advisor's accuracy directly. Our contributions are:

- A biologically grounded evaluation of context-dependent learning in VLMs, adapted from a human advice-taking paradigm, with text and image reports.
- Evidence that the failure lies in judging outcomes: models given raw histories repeat the previous outcome, a shortcut human participants did not use. **[TODO: confirm on held-out]**
- A comparison with humans in a second task, the changing bandit of Psych-101, with calibrated power.
- An open benchmark: task generator, prompts and analysis code; third-party data are fetched by script. **[TODO: anonymous code link]**

Related Work

Cognitive evaluation of LLMs. Treating a language model as a participant in a psychological experiment is now a common way to characterize its behavior (Binz and Schulz 2023). CogBench turns such experiments into behavioral metrics (Coda-Forno et al. 2024), and Centaur was fine-tuned to predict trial-by-trial human choices in Psych-101

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

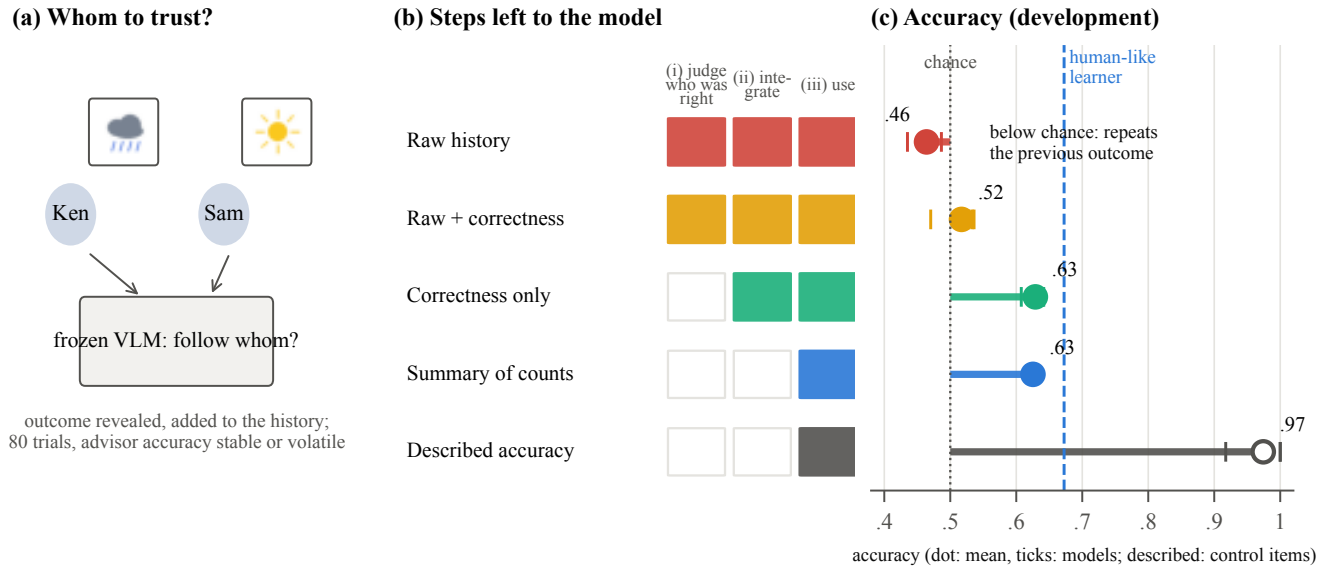


Figure 1: VLMs fail to learn whom to trust from raw feedback, and the failure lies in judging outcomes. (a) Two advisors give conflicting reports; after the model chooses whom to follow, the outcome is revealed and added to the history. (b) Each history format leaves different processing steps to the model (filled). (c) Accuracy with that format: mean over models (dot) and individual models (ticks); the dashed line is a Bayesian learner of the kind that describes human participants, run on the same episodes. Described accuracy is a control condition with its own items. **[DRAFT: development episodes (16) and four models; replace with held-out results]**

(Binz et al. 2025). Fits of reinforcement-learning models to in-context behavior in bandits have found asymmetric or biased updating and rigid or greedy choice (Schubert et al. 2024; Hayes, Yax, and Palminteri 2024; Wang, Ward, and Zhang 2026b; Schmied et al. 2025), and reversal-learning schedules have been compared across LLMs (Wang, Ward, and Zhang 2026a; Li et al. 2025). In all of these tasks the model learns about its own options; it does not have to check another agent's report against an outcome.

Description, experience and history summaries. The description–experience gap (Hertwig et al. 2004; Wulff, Mergenthaler-Canseco, and Hertwig 2018) has an LLM counterpart in the description–history gap of Ge, Zhang, and Vorobeychik (2026). Krishnamurthy et al. (2024) contrasted raw and summarized histories in bandits, and Xue, Wilson, and Xiong (2026) traced integration kernels in a counting task. LLMs also update roughly as Bayesian observers when the observed sequence is itself what they predict (Gupta et al. 2025; Zhang, Yang, and Wang 2025). Our summarized history is computed from the same history the model would otherwise see; it discards trial order and is not a description.

Volatility and multimodal reports. Our task follows the social-learning paradigm of Behrens et al. (2008). Normative accounts of learning under volatility include hierarchical Bayesian and change-point models (Behrens et al. 2007; Nassar et al. 2010, 2012; Mathys et al. 2011; Piray and Daw 2021, 2024; Soltani and Izquierdo 2019), and social and individual learning can be arbitrated separately (Diaconescu et al. 2020). For image reports, performance depends on modality even when content is isomorphic (Fu et al. 2024), multi-image inputs show position biases (Tian et al. 2025), and VLMs have difficulty binding features to objects (Campbell et al. 2024).

Task and Methods

Advice-taking task. On each trial, two advisors give conflicting binary reports (for example, rain and dry), the model predicts the outcome, which amounts to choosing whom to follow, and the outcome is then revealed. An episode has 80 trials. Advisor A is correct with probability 0.8 in a stable phase of 40 trials; in a volatile phase of 40 trials its accuracy alternates between 0.8 and 0.2 every 10 trials, following Behrens et al. (2008) (comparison in Appendix B). Half of the episodes start with each phase. Outcome sequences are pre-generated, so the model's choices do not change the history, and within each 10-trial block A's realized accuracy equals the scheduled value. The instructions say that an advisor's accuracy may change but give no rate.

Each episode is shown with a photo cover story, in which reporters say whether a hidden animal is a cat or a dog, and with one of six text cover stories (colors, weather, stocks, sports, medical tests, news; Appendix A). Current reports are given as text or as an image: a drawn icon for the text cover stories and a photograph (Parkhi et al. 2012) for the photo cover story. Every episode has a counterbalanced copy in which advisor names and answer labels are swapped while outcomes are identical; averaging over the two cancels name

and label biases. There are 16 development and 48 held-out episodes, and a robustness set of 24 episodes with independent outcomes and irregular reversals.

History formats. Before trial t, the model sees trials $1, \ldots, t-1$ in one format (Fig. 1b; prompts in Appendix A). The *raw history* lists both reports and the outcome ("Day 3: Ken forecasts rain; Sam forecasts dry. Actual weather: rain."). *Raw + correctness* appends "(Ken was correct; Sam was wrong.)" to each trial. *Correctness only* keeps only that statement, removing step (i). The *summary* gives each advisor's number of correct trials so far, removing steps (i) and (ii); it discards trial order. Further summaries count over the last K trials or since a change point inferred by Bayesian online change-point detection (BOCPD; Adams and MacKay 2007; Appendix E), and a chart draws who was correct as an image.

Control conditions. On development episodes, control items give the answer more directly: *described accuracy* ("Ken was correct on 80% of past days"), described counts, short and long easy histories in raw and correctness-only form, and a single raw trial on which the model names who was correct (step (i) alone).

Models and readout. We test five open-weight VLMs, without fine-tuning, from two language-model families: Qwen3-VL 8B and 32B (Bai et al. 2025), InternVL3.5 8B and 38B, which use a Qwen3 language model (Wang et al. 2025), and Gemma-3-27B (Gemma Team 2025). Two proprietary models are tested through their APIs on a subset (Appendix K). **[TODO: API runs]** All models run without reasoning and answer with one word. The readout is the choice probability y_t for advisor A's report, from the logits of the two answer words at the first generated token, normalized over these two words; it is not a belief. Nearly all probability mass fell on the two answer words (Appendix K).

Measures and analysis. Accuracy is the proportion of trials ($t \geq 2$) on which the followed advisor was correct, averaged over the counterbalanced copies; chance is 0.5. To describe behavior we fit a logistic regression of y_t on which advisor was correct k trials back, whose weights over k form an integration kernel (cf. Xue, Wilson, and Xiong 2026), plus a regressor for repeating the previous outcome, and compute agreement (Cohen's κ) with simple strategies. We also fit learning models (a constant bias, delta rules, counts over all or recent trials, change-point and volatility learners) to each model's choice probabilities and compare them by leave-one-episode-out cross-validation (Appendix F). As a reference we run the Bayes-optimal volatility learner that Behrens et al. (2007, 2008) fitted to their participants on the same episodes. The unit of analysis is the episode pair (an episode and its counterbalanced copy), and uncertainty is reported as 95% confidence intervals (CIs) from a bootstrap over episode pairs (10,000 resamples). The primary test, fixed before held-out evaluation, is the summary−raw accuracy difference per model (Wilcoxon signed-rank test, Holm-corrected; required in both families). Secondary tests and all other details are in Appendix C.

Results

[DRAFT: development episodes (16) and four models; replace with held-out results]

Models use described accuracy, but not experienced reliability. With text reports, described accuracy or counts led to the better advisor on 92–100% of items, and every model named the correct advisor on a single shown trial (Fig. 2b). The same information was used much less when it had to be learned. In an easy raw history in which one advisor had been right on every recent trial, models followed that advisor on .60–.70 of items after 10 trials and on .53–.56 after 80. Stating who was correct on each trial raised this to nearly 1.00 for 10 trials, but only to .68–.85 for 80 trials. Judging one trial is therefore not the difficulty; judging many trials is, and long histories remain hard even when this step is done for the model.

Stating who was correct closes the gap; appending it does not. On the main task, accuracy was .435–.487 with raw histories and .616–.632 with summaries (Fig. 2a; Table 1). The gap held in both language-model families, including Gemma-3-27B. A correctness-only history reached the level of the summary (.608–.642) and recovered 91–120% of the gap, although it still leaves integration to the model. Integration is thus not where most accuracy is lost. Appending the same statement to the raw history recovered only one-fifth to two-fifths of the gap, even though the model then had all the information of the correctness-only history. The raw reports and outcomes therefore do more than add work: they interfere with using the statement next to them, which the next analyses explain.

Under raw histories, models repeat the previous outcome. The kernel weights on past correctness were near zero, while the weight on repeating the previous outcome was positive in every model (Fig. 3a). Repeating the previous outcome agreed best with the models' choices ($\kappa = .24$–$.35$; Fig. 3b). Learning models that track the advisors explained raw-history choices barely better than a constant bias (at most 0.017 nats per trial), whereas with correctness-only or summarized histories they improved the fit by 0.02–0.10 nats, and counting correct trials over all past trials was usually the best model (Appendix F). Because each 10-trial block contains each outcome five times, copying scores below chance (.439), which explains accuracy below .5. When the history stated who was correct, the kernel instead peaked at the most recent trial and decayed over the next few, and the weight on the previous outcome vanished. To find what triggers the shortcut, we replaced parts of the raw history with correctness by a dash while keeping its layout and length. Replacing the outcome label raised accuracy by 7–10 points in the two 8B models and largely removed the weight on the previous outcome, whereas replacing the reports did not (Fig. 3c; Appendix J). With both replaced, accuracy came within about 2 points of the correctness-only history, so the length of the record explains little. In these models, the visible outcome label is what is copied.

A learner that describes people solves the same episodes. The Bayesian volatility learner fitted to Behrens et al.'s par-

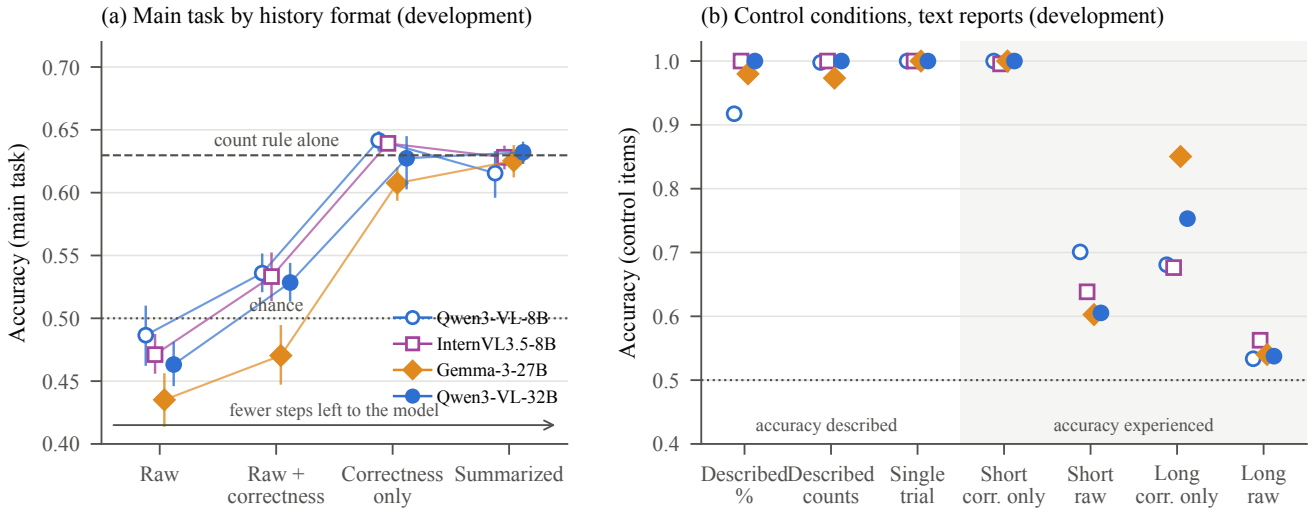


Figure 2: Models use described accuracy but not raw experience. (a) Main-task accuracy by history format, one line per model (shape and colour: family; filled: larger models; bars: 95% bootstrap CIs over episode pairs). (b) Control items: reliability given (left) or experienced (shaded). **[DRAFT: development episodes (16) and four models; replace with held-out results]**

ticipants, run on the same development episodes, chose the better advisor on .672 of trials, above every model in every format and above the count rule (.630; Appendix B). Those participants did not treat the advisor as repeating its last behaviour (Behrens et al. 2008); our models instead repeated the last outcome. With 80 trials, even this learner's learning rate rises only slightly in the volatile phase, too little to test volatility adaptation here (Appendix F).

A summary passes on its rule, including its lag. Given all-trial counts, models chose the advisor with more correct trials on 93–99% of non-tied trials, and three of four matched the rule's accuracy within ± 2 points (Table 2). A model that follows a summary therefore inherits the rule's response to change. The all-trial count lags behind every reversal (Appendix A, Fig. 6b), and after a reversal, models given the summary or the correctness-only history first kept following the formerly better advisor and needed most of the 10-trial block to switch, whereas with raw histories following stayed near .50 throughout (Fig. 4). Summaries over the last K trials or since an inferred change point respond faster; whether models follow them as closely decides how an agent built this way adapts to a volatile source. **[TODO: change-point and window summaries (held-out)]**

Image reports add a binding problem (exploratory). Models named a single report image almost perfectly, yet with described accuracy and image reports they followed the better advisor on only .65–.84 of items, least when the advisor's name was given only after the image (Appendix G). Accuracy was higher when the name was printed on the image or both reports were shown in one labelled image, although each panel was read correctly. With the correctness-only history, image reports lowered main-task accuracy by about 9 points compared with text reports. Models could not read single cells of the correctness chart reliably (.60–.71), so chart histories are not interpreted.

Humans and models in the changing bandit. In the changing bandit of Psych-101, a normative learner forgets the unchosen option faster when the hazard rate is higher, and participants did (median forgetting rate .079 at $h = 0.1$, .255 at $h = 0.4$; Appendix I). Models were conditioned on the same participants' histories without being told the hazard rate. Unlike the advice task, this task varies volatility across games, so it tests whether a learner adapts its forgetting to the environment. We compare each model's slope of forgetting rate on hazard rate with that of the participant whose history it saw, and repeat the comparison with the hazard rate stated. Participants had also played earlier games, so the comparison is descriptive. A simulation of the whole pipeline gave a false-positive rate of 6% but only about half the power for a difference as large as the human effect (Appendix I), so a null result would not show similarity (Fig. 5). **[TODO: model estimates and CIs]**

Reasoning and proprietary models. [TODO: reasoning subset (Appendix H) and API models (Appendix K), one sentence each]

Discussion

[DRAFT: development episodes (16) and four models; replace with held-out results] The VLMs used described accuracy and followed a summary as faithfully as its rule, yet chose at or below chance when they had to learn from the raw record. The loss lies in judging outcomes: once each trial stated who was correct, accuracy matched the summary although integration was still left to the model. This refines the description–history gap (Ge, Zhang, and Vorobeychik 2026): the gap between described and experienced accuracy

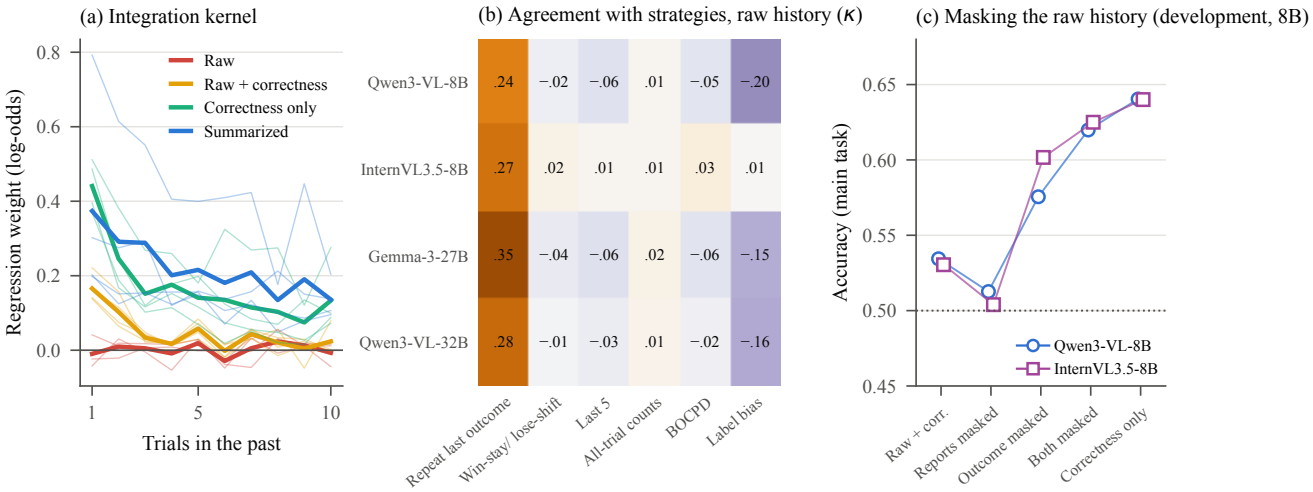


Figure 3: Under raw histories, models repeat the previous outcome. (a) Integration kernel by history format (thick: mean over models; thin: individual models; per-model 95% CIs in Appendix F). (b) Agreement with simple strategies under raw histories (Cohen's κ). (c) Replacing the outcome label in the raw history with correctness, but not the reports, recovers most of the gap to the correctness-only history (two 8B models, development episodes; points are model means). **[DRAFT: development episodes (16) and four models; replace with held-out results]**

is not mainly a failure to integrate experience but to turn it into outcomes worth integrating, which also explains why summaries help in bandit tasks (Krishnamurthy et al. 2024). In the human study, this step corresponds to a prediction error about the advisor, found in dorsomedial prefrontal and superior temporal cortex, while the advice's volatility was tracked in the anterior cingulate gyrus (Behrens et al. 2008); the models fail at forming this error, not at integrating it once it is given.

Implications for NeuroAI. For building agents, the results suggest the division of labour described in people: an explicit step that records which source was correct, and an integration rule outside the model whose time scale matches the expected volatility (Appendix D). For neuroscience, the format decomposition tests candidate models of advice taking step by step against the same signatures as people; the VLMs tested here are not yet such models. These parallels are computational; we make no claim about neural mechanisms. **[TODO: update with held-out results, summaries, reasoning and the human comparison]**

Limitations. The task is binary, with conflicting reports and fixed outcome quotas, which allow shortcuts; we therefore repeat the primary test on episodes with independent outcomes. Eighty-trial episodes are too short to test volatility-dependent learning rates, which we test only in the changing bandit, where the human comparison is underpowered. The summary discards trial order, and the size of the raw–summary gap depends on the history's layout (Appendix J). We tested two open model families and one reasoning model on a subset; images enter only as current reports. The analysis plan was fixed by a timestamped commit; deviations are listed in Appendix K.

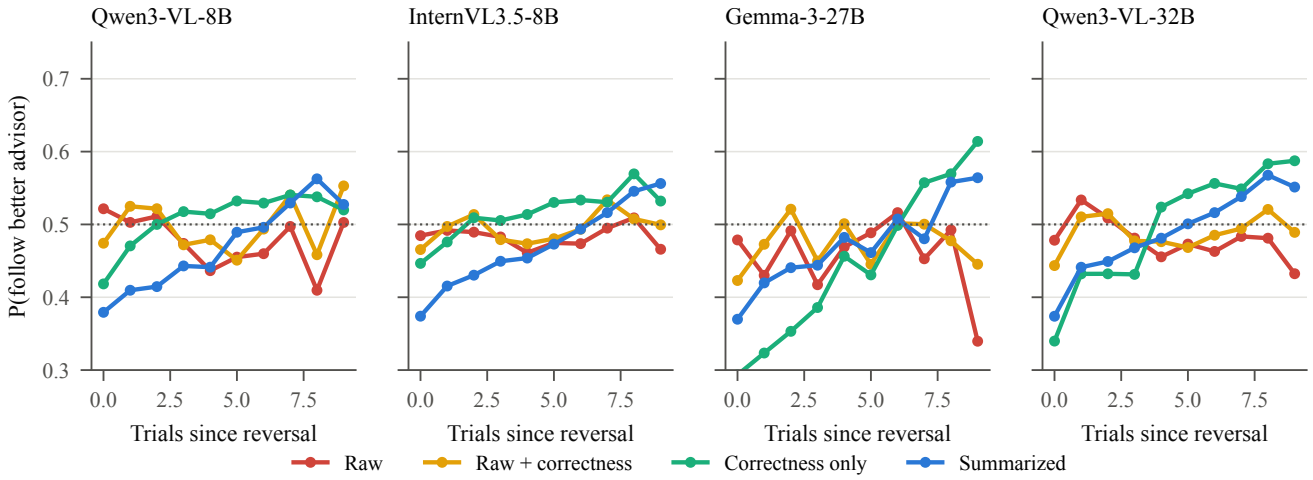


Figure 4: After a reversal, no format lets models switch quickly. Probability of following the currently better advisor as a function of trials since a reversal, by history format and model (mean over reversals and episodes; no error bars shown). **[DRAFT: development episodes (16) and four models; replace with held-out results] [TODO: add window and change-point summaries]**

Figure placeholder: Forgetting rate ϕ vs. hazard rate: humans (told h; our re-analysis of Xiong et al. 2023) and models (hidden h; told h as secondary).

Figure 5: **[TODO: Human comparison on Psych-101.]**

		Main-task accuracy (held-out, 48 episodes)						Described
Model	LM	Raw	Raw+corr.	Corr. only	Chart	Summary	Summary−raw [95% CI]	accuracy
Qwen3-VL-8B	Qwen3	.487	.536	.642	–	.616	+.129 [.105, .152]	.917
InternVL3.5-8B	Qwen3	.471	.533	.639	–	.628	+.157 [.141, .173]	1.000
Gemma-3-27B	Gemma 3	.435	.470	.608	–	.625	+.190 [.170, .211]	.980
Qwen3-VL-32B	Qwen3	.463	.528	.627	–	.632	+.169 [.152, .185]	1.000
InternVL3.5-38B	Qwen3	**??**	**??**	**??**	**??**	**??**	**??**	**??**
GPT-6-Astra (API)*	–	**??**	**??**	**??**	–	**??**	**??**	**??**
Claude Fable 5.2 (API)*	–	**??**	**??**	**??**	–	**??**	**??**	**??**
Bayesian learner (Behrens et al.)	–	.668 (raw outcomes; development: .672)					–	–
Count rule (all trials)	–	.632 (development: .630)					–	–

Table 1: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Raw history is at or below chance in every model; stating who was correct reaches the summary. Accuracy: proportion of trials ($t \geq 2$) on which the model follows the correct advisor, averaged over counterbalanced copies and episodes; text reports. LM: language-model family. Raw+corr.: raw history with who was correct appended; Corr. only: who was correct only; Chart: who was correct drawn as an image. Summary−raw: paired difference with 95% CI. Described accuracy: control items (development). *API models on a subset, choice read from the generated answer (Appendix K).

	Primary: summary − raw				Secondary: VLM+summary − rule alone (TOST, ±2 points)					
					All trials		Change-point		Selected window	
Model	Est.	95% CI	p	p_{Holm}	Δ [90% CI]	p_{TOST}	Δ [90% CI]	p_{TOST}	Δ [90% CI]	p_{TOST}
Qwen3-VL-8B	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**
InternVL3.5-8B	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**
Gemma-3-27B	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**
Qwen3-VL-32B	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**

Table 2: Primary and secondary tests on held-out episodes, fixed before held-out evaluation (timestamped commit; not a formal preregistration). Primary: Wilcoxon signed-rank test, Holm-corrected across models; the claim requires a significant positive gap in both families. Secondary: two one-sided tests (TOST) of equivalence within ±2 percentage points, per model, Holm-corrected over the three summaries; the window length is selected on development episodes (Appendix E).

References

Adams, R. P.; and MacKay, D. J. C. 2007. Bayesian Online Changepoint Detection. *arXiv preprint arXiv:0710.3742*.

Ashkinaze, J.; Kurek, L.; Faisal, A.; Miao, T.; Joseph, M.; Budak, C.; and Gilbert, E. 2026. Information Discernment in Large Language Models. arXiv:2607.19355.

Bai, S.; Cai, Y.; Chen, R.; Chen, K.; Chen, X.; Cheng, Z.; et al. 2025. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Behrens, T. E. J.; Hunt, L. T.; Woolrich, M. W.; and Rushworth, M. F. S. 2008. Associative learning of social value. *Nature*, 456(7219): 245–249.

Behrens, T. E. J.; Woolrich, M. W.; Walton, M. E.; and Rushworth, M. F. S. 2007. Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9): 1214–1221.

Binz, M.; Akata, E.; Bethge, M.; Brändle, F.; Callaway, F.; Coda-Forno, J.; Dayan, P.; Demircan, C.; Eckstein, M. K.; Éltető, N.; Griffiths, T. L.; Haridi, S.; Jagadish, A. K.; Ji-An, L.; Kipnis, A.; Kumar, S.; Ludwig, T.; Mathony, M.; Mattar, M.; Modirshanechi, A.; Nath, S. S.; Peterson, J. C.; Rmus, M.; Russek, E. M.; Saanum, T.; Schubert, J. A.; Schulze Buschoff, L. M.; Singhi, N.; Sui, X.; Thalmann, M.; Theis, F. J.; Truong, V.; Udandarao, V.; Voudouris, K.; Wilson, R.; Witte, K.; Wu, S.; Wulff, D. U.; Xiong, H.; and Schulz, E. 2025. A foundation model to predict and capture human cognition. *Nature*, 644(8078): 1002–1009.

Binz, M.; and Schulz, E. 2023. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6): e2218523120.

Bonaccio, S.; and Dalal, R. S. 2006. Advice taking and decision-making: An integrative literature review, and implications for the organizational sciences. *Organizational Behavior and Human Decision Processes*, 101(2): 127–151.

Campbell, D.; Rane, S.; Giallanza, T.; De Sabbata, N.; Ghods, K.; Joshi, A.; Ku, A.; Frankland, S. M.; Griffiths, T. L.; Cohen, J. D.; and Webb, T. W. 2024. Understanding the Limits of Vision Language Models Through the Lens of the Binding Problem. In *Advances in Neural Information Processing Systems*, volume 37, 113436–113460.

Chen, Y. 2026. Trust Between AI Agents: Measuring Formation, Breakage, and Recovery, with Implications for Governing Multi-Agent Systems. arXiv:2606.14923.

Coda-Forno, J.; Binz, M.; Wang, J. X.; and Schulz, E. 2024. CogBench: a large language model walks into a psychology lab. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 9076–9108. PMLR.

Diaconescu, A. O.; Stecy, M.; Kasper, L.; Burke, C. J.; Nagy, Z.; Mathys, C.; and Tobler, P. N. 2020. Neural arbitration between social and individual learning systems. *eLife*, 9: e54051.

Feng, P.; Yang, S.; and Poria, S. 2026. Σ-Mem: An Online Reliability Memory for LLM-based Multi-Agent Systems. arXiv:2607.27958.

Fu, D.; Guo, R.; Khalighinejad, G.; Liu, O.; Dhingra, B.; Yogatama, D.; Jia, R.; and Neiswanger, W. 2024. IsoBench: Benchmarking Multimodal Foundation Models on Isomorphic Representations. In *First Conference on Language Modeling (COLM)*. ArXiv:2404.01266.

Ge, L.; Zhang, Y.; and Vorobeychik, Y. 2026. Mind the (DH) Gap! A Contrast in Risky Choices Between Reasoning and Conversational LLMs. *arXiv preprint arXiv:2602.15173*.

Gemma Team. 2025. Gemma 3 Technical Report. *arXiv preprint arXiv:2503.19786*.

Gupta, R.; Corona, R.; Ge, J.; Wang, E.; Klein, D.; Darrell, T.; and Chan, D. M. 2025. Enough Coin Flips Can Make LLMs Act Bayesian. *arXiv preprint arXiv:2503.04722*.

Hayes, W. M.; Yax, N.; and Palminteri, S. 2024. Large Language Models are Biased Reinforcement Learners. arXiv:2405.11422.

Hertwig, R.; Barron, G.; Weber, E. U.; and Erev, I. 2004. Decisions from Experience and the Effect of Rare Events in Risky Choice. *Psychological Science*, 15(8): 534–539.

Krishnamurthy, A.; Harris, K.; Foster, D. J.; Zhang, C.; and Slivkins, A. 2024. Can Large Language Models Explore In-Context? In *Advances in Neural Information Processing Systems*. ArXiv:2403.15371.

Lakens, D. 2017. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, 8(4): 355–362.

Li, L.; Wang, Y.; Zhao, H.; Kong, S.; Teng, Y.; Li, C.; and Wang, Y. 2025. Reflection-Bench: Evaluating Epistemic Agency in Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 36236–36264. PMLR.

Mathys, C.; Daunizeau, J.; Friston, K. J.; and Stephan, K. E. 2011. A Bayesian foundation for individual learning under uncertainty. *Frontiers in Human Neuroscience*, 5: 39.

Nassar, M. R.; Rumsey, K. M.; Wilson, R. C.; Parikh, K.; Heasly, B.; and Gold, J. I. 2012. Rational regulation of learning dynamics by pupil-linked arousal systems. *Nature Neuroscience*, 15(7): 1040–1046.

Nassar, M. R.; Wilson, R. C.; Heasly, B.; and Gold, J. I. 2010. An approximately Bayesian delta-rule model explains the dynamics of belief updating in a changing environment. *Journal of Neuroscience*, 30(37): 12366–12378.

Parkhi, O. M.; Vedaldi, A.; Zisserman, A.; and Jawahar, C. V. 2012. Cats and dogs. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 3498–3505.

Piray, P.; and Daw, N. D. 2021. A model for learning based on the joint estimation of stochasticity and volatility. *Nature Communications*, 12: 6587.

Piray, P.; and Daw, N. D. 2024. Computational processes of simultaneous learning of stochasticity and volatility in humans. *Nature Communications*, 15: 9073.

Schmied, T.; Bornschein, J.; Grau-Moya, J.; Wulfmeier, M.; and Pascanu, R. 2025. LLMs are Greedy Agents: Effects of RL Fine-tuning on Decision-Making Abilities. arXiv:2504.16078.

Schubert, J. A.; Jagadish, A. K.; Binz, M.; and Schulz, E. 2024. In-Context Learning Agents Are Asymmetric Belief Updaters. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 43928–43946. PMLR.

Soltani, A.; and Izquierdo, A. 2019. Adaptive learning under expected and unexpected uncertainty. *Nature Reviews Neuroscience*, 20(10): 635–644.

Tian, X.; Zou, S.; Yang, Z.; and Zhang, J. 2025. Identifying and Mitigating Position Bias of Multi-image Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10599–10609.

Wang, H.; Ward, T. E.; and Zhang, L. 2026a. Comparative reversal learning reveals rigid adaptation in LLMs under non-stationary uncertainty. Accepted to IPMU 2026 (per arXiv comment), arXiv:2604.04182.

Wang, H.; Ward, T. E.; and Zhang, L. 2026b. Rigidity in LLM Bandits with Implications for Human-AI Dyads. arXiv:2603.07717.

Wang, W.; Gao, Z.; Gu, L.; Pu, H.; Cui, L.; Wei, X.; et al. 2025. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency. *arXiv preprint arXiv:2508.18265*.

Wulff, D. U.; Mergenthaler-Canseco, M.; and Hertwig, R. 2018. A meta-analytic review of two modes of learning and the description-experience gap. *Psychological Bulletin*, 144(2): 140–176.

Xiong, H.-D.; Ji-An, L.; Mattar, M. G.; and Wilson, R. C. 2023. Neural network modeling reveals diverse human exploration behaviors via state space analysis. In *Conference on Cognitive Computational Neuroscience (CCN)*.

Xue, J.; Wilson, R. C.; and Xiong, H. 2026. Counting on Thinking: Tracing Evidence Integration in Language Models. *arXiv preprint arXiv:2609.32932*.

Yaniv, I.; and Kleinberger, E. 2000. Advice Taking in Decision Making: Egocentric Discounting and Reputation Formation. *Organizational Behavior and Human Decision Processes*, 83(2): 260–281.

Zhang, J.; Yang, J.; and Wang, K. 2025. Large Language Models as Discounted Bayesian Filters. *arXiv preprint arXiv:2512.18489*.

Zhou, R.; Song, M.; Wu, X.; Cheng, S.; Yin, X.; Xie, Y.; Hao, Z.; Hua, W.; Pan, L.; Poria, S.; and Kan, M.-Y. 2026. Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems. ICLR 2026 Workshop on Agentic AI in the Wild, arXiv:2601.21742.

Supplementary Material

A. Prompts and Cover Stories

Every prompt has four parts: instructions naming the two advisors and the cover story, the history in one of the formats of the main text, the current reports, and the question. Prompts are passed through each model's chat template as a single user message. For InternVL3.5 the template adds the model's default system message (in Chinese) and the suffix "Answer the question using a single word or phrase." All answer words are single tokens in every tokenizer used.

Task details. Each episode is shown with two cover stories that share the same outcome sequence: a photo cover story, in which reporters say whether a hidden animal is a cat or a dog, and one of six text cover stories (colors, weather forecasts, stock predictions, sports predictions, medical readings, news claims), assigned in blocks so that cover story is crossed with phase order (Fig. 6). Cover stories change only the wording, the answer words and the report images; all outcome sequences are synthetic. Current reports are text or an image: a programmatically drawn icon without text for the text cover stories, and a photograph from the Oxford-IIIT Pet dataset (Parkhi et al. 2012) for the photo cover story. Both advisors' correctness is revealed on every trial (full feedback). In the counterbalanced copy, advisor names, answer labels and the order in which the instructions introduce the advisors are swapped while outcomes and images are identical; within each trial the two reports are listed in random order. The instructions say that an advisor's accuracy may change over time but give no rate. The held-out set has eight episodes per text cover story.

Table 3 lists the cover stories. The examples below are rendered by the experiment code from the fixed episode files; line breaks are as in the prompt and *[image]* marks where an image is inserted.

Cover story	Advisors	Answers	Report image
color	reporters	red/blue	color patch
weather	forecasters	rain/dry	cloud, sun
stock	analysts	up/down	trend arrow
sports	commentators	win/lose	happy/sad face
medical	doctors	positive/negative	plus/minus
news	news sources	true/false	tick, cross
pet (photo)	reporters	cat/dog	photograph

Table 3: Cover stories with the selected icon set (v3.1). Trials are called trials (color, pet), days, weeks, games, cases or stories. Icons are drawn by code and contain no answer words. Appendix G shows the original and selected medical and sports icons and identifies results obtained with the original set.

Example 1: raw history, weather, held-out episode, trial 6.

> You are observing a sequence of days. On every day, two weather forecasters, Tim and Joe, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
> Past days:
> Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 2: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Day 3: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.

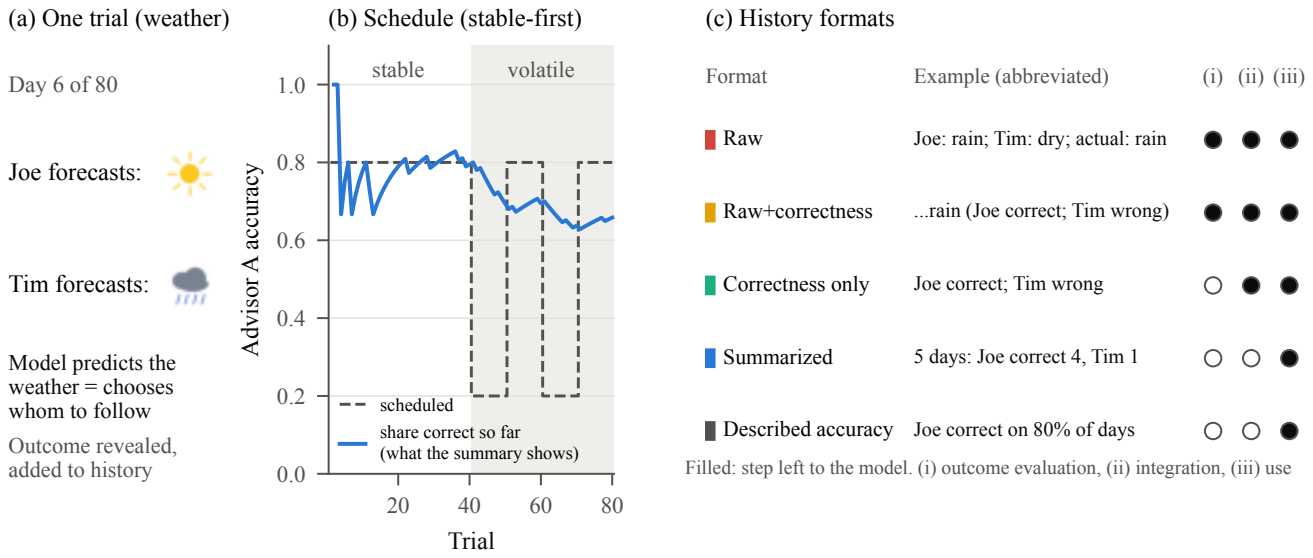


Figure 6: Task and history formats. (a) One trial in the weather cover story. (b) Scheduled accuracy of advisor A in a stable-first episode (dashed) and the share of past trials on which A was correct, which the all-trial summary reports (solid): it tracks the stable phase but lags behind every reversal. (c) History formats and the processing steps each leaves to the model: (i) outcome evaluation, (ii) integration across trials, (iii) use.

> Day 4: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 5: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Current day 6 (actual weather not yet observed):
> Joe forecasts: dry
> Tim forecasts: rain
> Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

The other formats change only the history lines. In the raw history with correctness, day 1 reads "Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry. (Joe was wrong; Tim was correct.)". In the correctness-only history it reads "Day 1: Joe was wrong; Tim was correct.". In the summarized history the five history lines and the "Past days:" line are replaced by "Summary of all 5 past days: Joe was correct 0 times; Tim was correct 5 times.". In the correctness chart they are replaced by "Past days 1-5 are summarized in this chart (days run left to right in four lines of 20; for each day, a green square means that forecaster was correct, a gray square means that forecaster was wrong, an empty square means not yet played):" followed by the chart (448 px; one row per advisor in each line). In the judgment readout the current reports and question are replaced by "Before day 6: which forecaster is currently more likely to be correct? Answer with exactly one name: Joe or Tim.".

Example 2: described accuracy, weather, development episode.

> You are observing a sequence of days. On every day, two weather forecasters, Jim and Ken, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
> Summary of past performance: Jim was correct on 80% of past days; Ken was correct on 20% of past days.
> Current day (actual weather not yet observed):
> Ken forecasts: rain
> Jim forecasts: dry
> Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

The described-counts control replaces the summary line with "Summary of the last 10 days: Jim was correct 8 times; Ken was correct 2 times.".

Example 3: photo cover story with image reports, held-out episode, trial 4.

> You are observing a sequence of trials. On every trial, two reporters, Jon and Jim, give conflicting reports about a hidden animal. After each trial, the true animal is revealed. A reporter's accuracy may change over time.
> Past trials:
> Trial 1: Jon reports cat; Jim reports dog. True animal: cat.
> Trial 2: Jon reports cat; Jim reports dog. True animal: cat.
> Trial 3: Jim reports dog; Jon reports cat. True animal: cat.
> Current trial 4 (true animal not yet revealed):
> Jim reports: *[image]*
> Jon reports: *[image]*
> Predict the true animal on the current trial. Answer with exactly one lowercase word: cat or dog.

With text reports the images are replaced by the answer word.

Binding variants. The binding controls change only how the current reports of Example 3 are attached to the advisors (described accuracy, image reports). *Name on image*: the lines are as in Example 3, but each image carries its advisor's name printed in a banner above the picture. *Composite image*:

the two report lines are replaced by "Reports (one panel per reporter, name above each panel): *[image]*", one image with two panels, each headed by an advisor's name. *Name after image*: each report is given as "*[image]* (the image above was reported by Jim)", so the attribution follows the image. The composite-read check shows only the composite image and asks "The image has two panels, each with a name above it. What animal is in Jim's panel?".

B. Episode Generation and Counterbalancing

Schedules. Each episode has 80 trials in eight blocks of 10. Advisor A is the advisor favored on trial 1. Stable-first episodes have a stable phase (A correct with probability 0.8) followed by a volatile phase with blocks at 0.2, 0.8, 0.2, 0.8; volatile-first episodes have a volatile phase with blocks at 0.8, 0.2, 0.8, 0.2 followed by a stable phase at 0.8. Every episode therefore has four reversals (trials 41, 51, 61, 71 in stable-first and 11, 21, 31, 41 in volatile-first episodes). Advisor B is always correct when A is wrong. For comparison, the advice schedule of Behrens et al. (2008) (Supplementary Information) has 120 trials: 75% correct on trials 1–30, alternation between 80% correct and 80% incorrect every 10 trials on trials 31–70, and 85% incorrect on trials 71–120. Our volatile phase matches their alternation; our stable phase does not.

Relation to the original human study. Table 4 compares our task with that of Behrens et al. (2008). Their 24 participants chose between two options with randomly generated reward magnitudes, saw a confederate's advice before choosing, and learned both the probability that each option was correct and how often the advice was right; both schedules had stable and volatile periods that did not coincide. A logistic regression showed that choices used reward magnitudes, the reward-based probability and the advice-based probability, each significantly, and to a similar extent, whereas following the confederate blindly or assuming that it would repeat its previous behaviour had no significant effect. A hierarchical Bayesian learner that tracks volatility (Behrens et al. 2007) described the learning, with faster learning when the source was volatile. We keep the core of the social part, an advisor whose accuracy is stable or alternates between 0.8 and 0.2 every 10 trials and is learned from revealed outcomes, and remove what a model without fine-tuning cannot be given or does not need: reward magnitudes, a second, reward-based learning stream, a confederate introduced in person, and the 120-trial length. Two advisors whose reports always conflict make the advisor's accuracy the only thing to learn. Because the original data are not public, we compare models with the learner that describes the participants, run on our episodes (Table 4, last rows), and with human data from Psych-101 (Appendix I).

Stratification. Within each block, A is correct on exactly $10p$ trials (8 or 2) in random order, and each outcome label is the truth on exactly five trials. This keeps realized accuracy equal to the schedule in every episode. It also fixes outcome quotas, which allow a shortcut that ignores the advisors: predicting the label with more occurrences remaining in the current block's quota (which presupposes knowing the quota) reaches 0.661 accuracy on the development episodes. We therefore also use a robustness set (below).

Names, order and counterbalancing. Advisor names are drawn from 16 common English first names that are single tokens in all tokenizers. The order in which the two reports are listed is random on each trial. Phase order alternates across episodes, the order in which the instructions introduce the two names alternates in pairs, and text cover stories rotate in blocks of four episodes, so cover story, phase order and introduction order are fully crossed. Each episode has a counterbalanced copy with the advisor names swapped (A keeps its role), the answer-label mapping reversed and the introduction order reversed; outcomes and images are identical. Analyses average the label-aligned logits within each pair.

Splits and seeds. The main set has 16 development episodes (color, weather, stock and sports cover stories) and 48 held-out episodes (eight per text cover story), 128 with counterbalanced copies, generated with seed 20261111. Pet photos come from the Oxford-IIIT Pet dataset (CC BY-SA 4.0): 128 photos from the official trainval split for development episodes and 256 from the test split for held-out episodes, with no overlap. The robustness set has 24 held-out episodes (48 with copies; seed 20261112) in which correctness is drawn independently on each trial, outcome labels are drawn independently, and volatile-phase dwell times are drawn uniformly from 6–16 trials. All data are regenerated from seeds and checked against stored SHA-256 hashes.

C. Statistical Analysis, Control Conditions and Decomposition

Statistical analysis. The unit of analysis is the episode pair; all measures are averaged within an episode and its counterbalanced copy first, after averaging the label-aligned logits, which cancels additive name and label biases. Besides accuracy we report the probability of following the currently more accurate advisor by phase, the log score of the outcome, and learning curves aligned to reversals. The primary test, fixed before held-out evaluation (timestamped commit; not a formal preregistration), is the summarized minus raw history accuracy for text reports, per model, with a two-sided Wilcoxon signed-rank test and Holm correction across models; the decision criterion is a gap significantly above zero in both language-model families. Raw-history accuracy is called near chance only within 0.5 ± 0.03. Secondary tests are two one-sided tests (TOST; Lakens 2017) of whether a model given a summary matches the count rule alone, with equivalence bounds of ± 2 percentage points; the position of the correctness-only history between raw and summarized history, $(\mathrm{acc}_{\text{corr. only}} - \mathrm{acc}_{\text{raw}})/(\mathrm{acc}_{\text{summary}} - \mathrm{acc}_{\text{raw}})$, interpreted only when the summary–raw gap exceeds 0.05 and its CI excludes zero; and the primary test on the robustness set, as a separate family. CIs are 95% bootstrap intervals over episode pairs (10,000 resamples). Cover story is a random factor: we report per-story estimates and a sign test across the six text cover stories (smallest attainable two-sided $p = .031$). Development episodes are used only for setting choices such as the window length K; held-out episodes are

	Behrens et al. (2008)	This study
Agents	24 human participants (fMRI)	Five VLMs without fine-tuning
Choice	Blue or green option, each with a random reward magnitude	Which of two binary reports is correct (whose advice to follow)
Sources of information	Reward history of the options and one confederate's advice	Two advisors whose reports always conflict
Feedback	Correct option revealed after each trial	Outcome revealed after each trial
Advice schedule	120 trials; 75% correct, then 80%/20% alternating every 10 trials, then 15%	80 trials; 80% (stable, 40 trials) and 80%/20% alternating every 10 trials (volatile, 40 trials), both orders
Analysis	Logistic regression of choices; hierarchical Bayesian volatility learner; fMRI	Accuracy by history format; integration kernels; strategy agreement; the same Bayesian learner as reference
Behavioural result	Advice weighted by its learned accuracy; "advisor repeats its previous behaviour" not used	Raw histories: choices repeat the previous outcome; correctness weights near zero
Bayesian volatility learner (Behrens et al. 2007) on our episodes (raw outcomes, no VLM)		
Accuracy	.672 (development), .668 [.661, .674] (held-out), .660 (robustness); all-trial count rule: .630 / .632 / .594	
Implied learning rate	stable .100, volatile .115 (held-out): too small a difference to detect in 80-trial episodes	

Table 4: The original human study and ours. The original advice schedule is from the Supplementary Information of Behrens et al. (2008); its reward schedule is omitted. Accuracy of the Bayesian learner: proportion of trials ($t \geq 2$) on which it favours the advisor who is correct, with a 95% bootstrap CI over episodes.

run once. The kernel regression uses lags $k = 1, \ldots, 15$ plus regressors for repeating the previous outcome, a label bias and listing order; strategies for κ are repeating the previous outcome, win-stay/lose-shift on advisors (following the advisor correct on the previous trial; the CPT regressor of Behrens et al. 2008), counts over all or recent trials, BOCPD and a label bias. These analyses describe behavior consistent with a strategy and do not identify a mechanism.

Decomposition. On development episodes the correctness-only history recovered 91–120% of the summary–raw gap, and appending the correctness statement to the raw history recovered $+.035$ to $+.065$ (Table 7). Integration over long histories still limits accuracy in the volatile phase, where following the better advisor was near .50 in all formats (Fig. 4), as it is for the all-trial count rule itself. With correctness-only histories the kernel was recency-weighted and the weight on the previous outcome vanished; with appended correctness both components were present.

Control conditions. Table 6 gives each control condition with text and image reports, and Table 7 the stepwise differences between history formats on the main task. Image reports lowered accuracy in almost every control condition, including described accuracy (.65–.84 instead of .92–1.00); the one exception was Gemma-3-27B with a short raw history (.612 vs. .603). The image-report results use the selected icons (v3.1, Appendix G), which every model named correctly on every single-image item (photos: .986–1.00), so these drops reflect the use of image reports rather than recognition of a single image. **[DRAFT: development episodes (16) and four models; replace with held-out results]**

	Described		Short		Long		Single
Model	%	counts	raw	corr.	raw	corr.	trial
Qwen3-VL-8B	.917	.998	.701	1.00	.533	.681	1.00
InternVL3.5-8B	1.00	1.00	.638	.996	.562	.676	1.00
Gemma-3-27B	.980	.973	.603	1.00	.540	.850	1.00
Qwen3-VL-32B	1.00	1.00	.607	1.00	.545	.750	1.00
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**	**??**	**??**

Table 5: Control conditions on development episodes, text reports (accuracy on control items, averaged over counterbalanced copies; image reports in Appendix C). Described: accuracy given as a percentage or as counts. Corr.: correctness only. Short/long: an easy history in which one advisor is clearly better, given raw or correctness-only. Single trial: who was correct on one shown trial. Full results by cover story in Appendix C.

D. Normative Simulation

We ran two simulations without any VLM to check the design. Both score a rule by the proportion of trials ($t \geq 2$) on which the advisor it favours is correct; ties score 0.5.

Design simulation. The first simulation draws outcomes independently from the schedule (300 episodes per setting) and compares BOCPD (posterior predictive, $h \in \{1/8, 1/15, 1/30\}$), majority counts over the last $K \in \{3, 5, 7, 10, 15, 20, 30\}$ trials, counts over all past trials, and counts since the last true reversal. Across 20 settings (accuracy 0.65–0.85; phase lengths 40/40, 60/60 and 80/40; dwell 10 or 15 trials), the best BOCPD minus the best fixed window ranged from -0.004 to $+0.008$. For our setting (0.8; 40/40; dwell 10), accuracy was 0.687 for the best BOCPD ($h = 1/8$), 0.683 for $h = 1/15$, 0.690 for the best window ($K = 3$), 0.632 for counts over all trials and 0.739 for counts

Condition	Reports	Qwen3-8B	IVL-8B	Qwen3-32B	Gemma-27B	IVL-38B
Described %	Text	.917	1.00	1.00	.980	**??**
	Image	.835	.806	.656†	.647	**??**
Described counts	Text	.998	1.00	1.00	.973	**??**
	Image	.855	.779	.644†	.647	**??**
Short raw	Text	.701	.638	.607	.603	**??**
	Image	.629	.545	.504†	.612	**??**
Short corr. only	Text	1.00	.996	1.00	1.00	**??**
	Image	.940	.746	.683†	.683	**??**
Long raw	Text	.533	.562	.545	.540	**??**
	Image	.540	.518	.409†	.507	**??**
Long corr. only	Text	.681	.676	.750	.850	**??**
	Image	.598	.429	.477†	.603	**??**
Single trial	Text	1.00	1.00	1.00	1.00	**??**
Described %, name on image	Image	.821	.830	.759	.734	**??**
Described %, composite image	Image	.748	.842	.770	.824	**??**
Described %, name after image	Image	.596	.696	.549	.520	**??**
Composite read	Image	1.00	1.00	1.00	1.00	**??**
Chart read	Chart	.630	.688	.672	.604	.714

Table 6: Control conditions on development episodes (accuracy on control items, averaged over counterbalanced copies and pooled over cover stories). Lower block: binding variants of the described-accuracy condition with image reports (advisor name printed on the image; both reports in one composite image; name given after the image), reading one panel of the composite image, and reading one cell of the correctness chart. Qwen3: Qwen3-VL; IVL: InternVL3.5. Image reports use the selected icons (v3.1). †Partial run (413 of 448 items per cell). Results by cover story are in the released analysis output.

Model	Raw+corr.	Corr. only	Chart	Summary−corr. only	Summary	Position
Qwen3-VL-8B	+.049	+.155	–	−.026	+.129	1.20 [1.08, 1.37]
InternVL3.5-8B	+.062	+.168	–	−.011	+.157	1.07 [1.02, 1.13]
Gemma-3-27B	+.035	+.172	–	+.018	+.190	0.91 [0.85, 0.96]
Qwen3-VL-32B	+.065	+.164	–	+.005	+.169	0.97 [0.85, 1.08]
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**	**??**

Table 7: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Decomposition of the raw–summarized gap: paired differences in main-task accuracy relative to raw history (except Summary−correctness only) with 95% bootstrap CIs (format: mean [low, high]). Position: (correctness only − raw)/(summary − raw), reported only when the summary–raw gap exceeds the prespecified threshold and its bootstrap CI excludes 0; otherwise "n/i".

since the last reversal. Conclusions were the same for accuracies from 0.65 to 0.85, and accuracy varied by about one percentage point for hazard rates between 1/8 and 1/30.

Rules given to the VLM. The second simulation applies exactly the rules used to build the summarized histories (Appendix E) to the actual episodes. Table 8 gives the results. On held-out episodes, the 5-trial window (.692 [.685, .698]) is about 1.4 percentage points above the change-point summary (.678 [.666, .688]); counts over all trials (.632) are 6.0 points below the 5-trial window and 4.6 points below the change-point summary. On the robustness set, the 5-trial window (.675) again leads the change-point summary (.663) and the reversal-aligned reference is highest (.735), but the remaining rules are ordered differently: on held-out episodes counts over all trials (.632) exceed the 10-trial (.612) and 20-trial (.579) windows, whereas on the robustness set the 10-trial (.635) and 20-trial (.604) windows exceed counts over all trials (.594). These differences hold for the tested settings only; we do not claim that online rules are equivalent in general. The reversal-aligned summary uses only reversals up to $t-1$ and so learns of a reversal one trial late; it is a reference, not an upper bound on any strategy. Likewise, 0.8 is the accuracy of an observer who knows the current accuracy, not a strict ceiling (Appendix B).

Summary rule	Development	Held-out	Robustness
All trials	.630	.632	.594
Last 5 trials	.698	.692	.675
Last 10 trials	.620	.612	.635
Last 20 trials	.578	.579	.604
Change-point (BOCPD)	.678	.678	.663
Since last reversal	.739	.744	.735

Table 8: Accuracy of the summary rules alone (no model) on development (16), held-out (48) and robustness (24) episodes. 95% bootstrap CIs over episodes (2,000 resamples), held-out: last 5 trials [.685, .698]; change-point [.666, .688]; all trials [.628, .636].

These rule baselines were computed on held-out episodes before any model results on the main data were seen. They are properties of the episode generator and involve no model behavior.

Baseline (no model)	Held-out	Robustness
Always follow A	??	??
Known current accuracy	??	??
BOCPD posterior	??	??
Outcome quota (knows block quota)	??	??

Table 9: Additional model-free baselines on the trial mask used for the model runs (95% CIs in the analysis output).

E. Change-Point Controller and Summary Variants

Change-point detection. Let $o_t = 1$ if advisor A was correct on trial t. Before trial t, the controller runs Bayesian online change-point detection (Adams and MacKay 2007) on $o_1, \ldots, o_{t-1}$ with a Bernoulli likelihood, a Beta(1,1) prior for each run and a constant hazard rate $h = 1/15$. This rate lies between the reversal rates of the volatile (1/10) and stable phases and was not tuned (Appendix D). The controller takes the maximum a posteriori run length r, sets it to at least 3, and summarizes trials $\max(1, t-r)$ to $t-1$.

Summary variants. All variants use the same sentence as the summarized history, restricted to a range of trials ("Summary of the last n days (days s–$(t-1)$): Ken was correct x times; Sam was correct y times."). The variants are: all past trials (the primary summary); the last $K \in \{5, 10, 20\}$ trials; the run since the last inferred change-point (BOCPD); and the trials since the last true reversal up to $t-1$ (reversal-aligned reference). Two further conditions truncate the raw history instead of summarizing it: to the BOCPD run, or to the last 10 trials. The window length used in the secondary tests is, for each model, the one with the highest accuracy on development episodes, fixed before held-out evaluation; Table 10 reports the selected length and all variants, and Table 11 compares the model with the rule alone.

What the controller is. The controller is a training-free rule outside the model; it sees only the outcome sequence and does not read model states. It is motivated by change-point models of human learning (Nassar et al. 2010, 2012), but we use it as an integration rule, not as a model of the VLM or of the brain.

F. Computational Models and Parameter Recovery

Models. Let $o_t = 1$ if advisor A was correct on trial t and y_t the model's choice probability for A. Each computational model maps $o_1, \ldots, o_{t-1}$ to a belief q_t that A is correct and predicts $P_t = \sigma(\beta(q_t - 0.5) + b)$, with inverse temperature β and bias b. M0 is a constant bias ($P_t = \sigma(b)$). Win-stay/lose-shift follows the advisor who was correct on the previous trial ($P_t = \sigma(b + w(2o_{t-1} - 1))$). Count models use $q_t = (s+1)/(n+2)$ over all past trials or over the last k trials. M1 is a delta rule, $q_{t+1} = q_t + \alpha(o_t - q_t)$ with $q_1 = 0.5$; M2 uses separate learning rates in the stable and volatile phases; M3 uses separate rates for outcomes that confirm or disconfirm the currently trusted advisor. M4 comprises a Bayesian change-point observer (hazard rate from a grid, or marginalized over it) and a volatility-tracking learner after Behrens et al. (2007), each either with a fitted readout or reading out its predictive probability directly.

Fitting and comparison. Because y_t is a choice probability rather than a sampled choice, models are fitted by minimizing the soft cross-entropy $-\sum_t [y_t \log P_t + (1 - y_t) \log(1 - P_t)]$, a pseudo-likelihood. Models are compared by leave-one-episode-out cross-validation: parameters are fitted jointly on all other episodes of a model and history format and scored on the left-out episode. Information criteria are reported only as heuristics. Following the analysis plan, learning-rate analyses are run only for conditions in which a model can use reliability at all (described accuracy ≥ 0.8 in the control conditions) and tracks the better advisor in the stable phase (probability ≥ 0.6); on development episodes, stable-phase following under raw histories was .53–.57 in every model, below this criterion, whereas it was .86–.98 with correctness-only and summarized histories.

Effective learning rate. The update of the model's choice probability in logit space after each outcome, regressed on the signed outcome separately for the stable and volatile phases, measures how strongly the model responds to new outcomes. This statistic is not zero for a learner with a single constant learning rate: preferences are more extreme in the stable phase, where the logit stretches updates. We therefore calibrate it against a null in which M1, fitted to the same episode, generates surrogate preferences with the same residual noise (200 surrogates); the calibrated volatile-minus-stable difference and its z score against the null indicate whether a model updates more in the volatile phase than a constant-rate learner would.

Recovery. In simulations with M1–M3 (100 synthetic learners each, logit noise s.d. 0.3), learning rates were recovered with correlations of .998 (M1; M2 stable and volatile rates) and .89–.99 (M3). Model identification was asymmetric: data from M1 were always classified as M1, but data from M2 were classified as M2 in 40% of cases (70% when the two rates differed by more than 0.2) and data from M3 rarely as M3 (10%). The sign of the M2 rate difference was recovered in every case with a clear difference. Phase-specific rates are thus estimable, whereas model comparison favors the simpler M1 unless the difference is large; we therefore base conclusions about volatility on the calibrated effective learning rate rather than on model selection alone. Table 14 checks the strategy and kernel analyses of the main text on synthetic agents run on the same episodes.

Results on development episodes. Leave-one-episode-out cross-validation (soft cross-entropy per trial) on the four models' development runs: with raw histories the best learning model improved on a constant bias by 0.001–0.017 nats per trial (values 0.66–0.69, near $\ln 2 = 0.693$), so no advisor-tracking model describes these choices; the repeat-the-previous-outcome strategy, which is not a learning model

		Summarized history						Truncated raw history	
Model	Selected K	All trials	Last 5	Last 10	Last 20	Change-point	Since reversal	Change-point	Last 10
Qwen3-VL-8B	??	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??	??
Rule alone	–	.632	.692	.612	.579	.678	.744	–	–

Table 10: Main-task accuracy for every summary variant and the truncated raw histories (held-out). Rule alone: the summary's majority rule without a model (Table 8).

Model	Summ.	VLM	Rule	Δ [90% CI]	Agree
Qwen3-VL-8B	All	.616	.630	$-.014$ [$-.026$, $-.004$]	.926
	CP	??	??	??	??
	Selected	??	??	??	??
InternVL3.5-8B	All	.628	.630	$-.002$ [$-.007$, .004]	.975
	CP	??	??	??	??
	Selected	??	??	??	??
Gemma-3-27B	All	.625	.630	$-.004$ [$-.011$, .003]	.946
	CP	??	??	??	??
	Selected	??	??	??	??
Qwen3-VL-32B	All	.632	.630	+.002 [$-.002$, .007]	.994
	CP	??	??	??	??
	Selected	??	??	??	??
InternVL3.5-38B	All	??	??	??	??
	CP	??	??	??	??
	Selected	??	??	??	??

Table 11: **[DRAFT: development episodes (16) and four models; replace with held-out results]** VLM with summarized history vs. the summary's rule alone on identical trials. Agree: proportion of non-tied trials on which the model's choice equals the rule's. Equal accuracy does not imply the same strategy; agreement is reported for that reason.

of advisor accuracy, is described by the kernel and strategy analyses instead. With the raw history with correctness the improvement was 0.009–0.030, with correctness-only histories 0.024–0.104 and with summaries 0.05–0.06 nats per trial. Counting correct trials over all past trials was the best model in 9 of the 12 non-raw cases (otherwise a delta rule or phase-specific or confirmation-biased delta rule), so when outcomes are judged for them, models integrate roughly as a running count. Because the calibrated effective learning rate cannot detect volatility adaptation on these episodes (below), we do not interpret its values. **[DRAFT: development episodes (16) and four models; replace with held-out results]**

Can these episodes reveal volatility-dependent learning? We checked this with the Bayesian volatility learner of Behrens et al. (2007), which does learn faster when accuracy is volatile. On our 80-trial episodes its implied learning rate rose only from .100 in the stable to .115 in the volatile phase (held-out episodes). Neither the calibrated volatile-minus-stable effective learning rate ($+0.002$ [-0.010, 0.014] over 16 development episodes) nor phase-specific integration kernels (volatile/stable ratio of the summed weight of the last three outcomes: 0.84–0.90, versus 0.96 for a constant-rate learner) detected this change, so we do not use the advice task to test whether models adapt their learning rate to volatility. The changing bandit (Appendix I), in which volatility differs across whole games, tests that question instead.

G. Icon Design, Multimodal Binding and Chart Legibility

The legibility and binding controls use development episodes only and are exploratory; the comparison of text and image reports on the main task uses held-out episodes. All use the selected icons (v3.1) described below. Every model named every single report image correctly (InternVL3.5-8B: .986 for pet photos), so the remaining costs are not recognition failures (Table 15). With described accuracy, accuracy was lowest when the advisor's name was given only after the image (.52–.70) and higher when the name was printed on the image (.73–.83) or both reports were shown in one composite image (.75–.84), although each panel of a composite image was read correctly on every item. With correctness-only histories, image reports lowered main-task accuracy from .608–.642 to .529–.553. Reading a single cell of the correctness chart was below the 0.8 gate for every model (.60–.71).

Icon design and examples. Icons replace the answer word with a picture in the image-report conditions. Figure 8 shows the actual deterministic renderings used in the icon probe, at

Model	History	Repeat last outcome	Win-stay/lose-shift	Last 5	All-trial counts	BOCPD	Label bias
Qwen3-VL-8B	Raw	0.24	−0.02	−0.06	0.01	−0.05	−0.20
	Summary	0.02	0.07	0.13	0.18	0.15	0.03
InternVL3.5-8B	Raw	0.27	0.02	0.01	0.01	0.03	0.01
	Summary	0.00	0.08	0.15	0.39	0.17	0.01
Gemma-3-27B	Raw	0.35	−0.04	−0.06	0.02	−0.06	−0.15
	Summary	0.00	0.07	0.15	0.25	0.15	0.02
Qwen3-VL-32B	Raw	0.28	−0.01	−0.03	0.01	−0.02	−0.16
	Summary	0.00	0.07	0.13	0.79	0.13	0.01
InternVL3.5-38B	Raw	??	??	??	??	??	??
	Summary	??	??	??	??	??	??

Table 12: **[DRAFT: development episodes (16) and four models; replace with held-out results]** Agreement (Cohen's κ) between the model's choices and simple strategies. κ corrects for chance agreement, so a rule that almost always picks one advisor can have high raw agreement and low κ. Repeat last outcome: predict the previous trial's outcome label. Win-stay/lose-shift: follow the advisor who was correct on the previous trial. Label bias: always predict the same label.

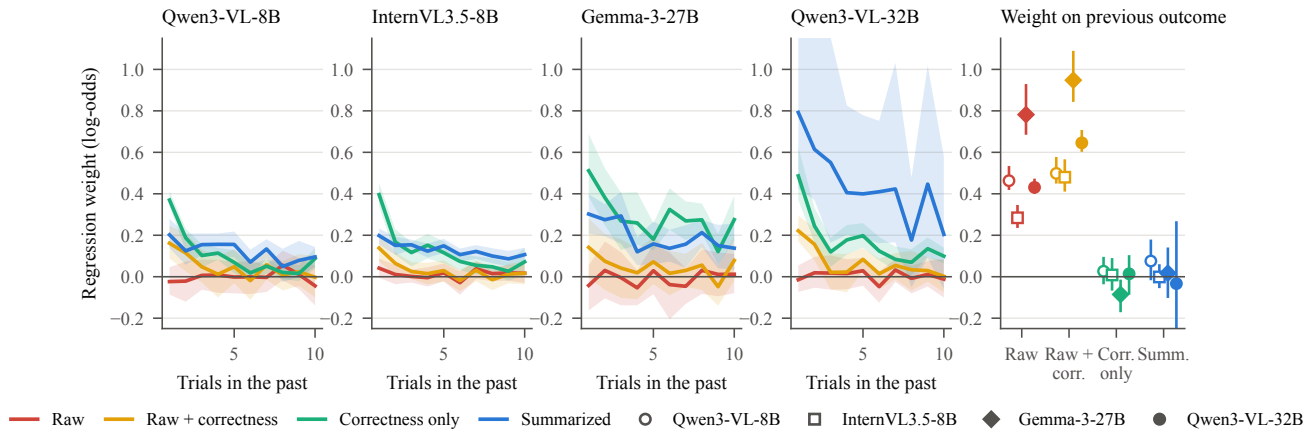


Figure 7: Integration kernels per model (shaded: 95% bootstrap CI over episodes) and, right, the weight on repeating the previous outcome by history format (bars: 95% CIs). **[DRAFT: development episodes (16) and four models; replace with held-out results]**

224 pixels; InternVL3.5 receives 448-pixel renderings from the same drawing functions. In the original medical pair, two red test-strip lines indicate positive and one indicates negative. The selected pair uses a dark plus or minus inside the same circular outline. In sports, the original gold trophy and red cross on a grey ball encode win and lose; the selected pair uses a yellow happy face and a grey-blue sad face. Labels and selection markers in the figure are reader annotations and are absent from the report images. These are synthetic symbols for the answer labels, not clinical scans or photographs of game outcomes. All other cover-story icons are unchanged.

Development-only selection. We tested five medical pairs and four sports pairs with Qwen3-VL-8B and InternVL3.5-8B before held-out evaluation. The single-image probe asks for each of the two labels under both answer orders (four prompts per pair and model). We required all four responses to be correct for both models, then selected the eligible pair with the highest mean accuracy over four image-report controls and both models (Table 16). The controls use described accuracy, a short raw history, a short correctness-only history, and a long correctness-only history. Each downstream cell contains 64 items after averaging label-aligned logits across the counterbalanced episode copies. This selected plus/minus and happy/sad faces. Enlarging the test strip left naming accuracy at .50 for both models, so size alone did not resolve that failure. Coloured plus/minus signs scored higher downstream but failed the InternVL3.5 naming criterion.

Results and limits. The downstream mean increases from .521 to .668 for medical and from .588 to .678 for sports. Both selected pairs reach 4/4 in the naming probe for both models. However, accuracy in the described-accuracy condition, pooled over these two image-report cover stories is .773 for Qwen3-VL-8B and .727 for InternVL3.5-8B, below the .80 gate. Single-image recognition therefore does not establish successful use of image reports in the advice-taking task. Selection and evaluation use the same development controls, so these gains are descriptive and do not estimate held-out improvement.

Model	History	Best	NLL/trial	Vol.−stable	z
Qwen3-VL-8B	Raw	??	??	??	??
	Summary	??	??	??	??
InternVL3.5-8B	Raw	??	??	??	??
	Summary	??	??	??	??
Gemma-3-27B	Raw	??	??	??	??
	Summary	??	??	??	??
Qwen3-VL-32B	Raw	??	??	??	??
	Summary	??	??	??	??
InternVL3.5-38B	Raw	??	??	??	??
	Summary	??	??	??	??

Table 13: Computational-model fits on held-out episodes: best model by leave-one-episode-out cross-validated soft cross-entropy per trial, and the calibrated volatile-minus-stable difference in effective learning rate with its z score against the constant-rate null.

Versioning and main-task check. This development-driven change is deviation D1 from the fixed plan. The selected icons define v3.1; v3 remains reproducible and original outputs are retained separately. Affected image-report and binding jobs require reruns. Text reports, pet photographs, the correctness chart, and the primary text-report comparison are unchanged. On the main task with correctness-only history and image reports (16 development episodes, both counterbalanced copies, trials $t \geq 2$), the selected icons raised mean accuracy over the original ones for both models: Qwen3-VL-8B from .51 to .62 (medical) and from .56 to .67 (sports), and InternVL3.5-8B from .50 to .55 and from .52 to .55 (paired differences over 32 episode copies: $+.11$ [.06, .16], $+.11$ [.09, .14], $+.05$ [.03, .06], $+.03$ [.02, .05]; 95% bootstrap CIs). Accuracy remained below the same condition with text reports (Qwen3-VL-8B .57 and .70; InternVL3.5-8B .79 and .77), so image reports still add a cost beyond outcome evaluation, now with icons that both models name correctly. The selected icons also generalized beyond the two models used to choose them: Qwen3-VL-32B, InternVL3.5-38B and Gemma-3-27B named them correctly on every item. Tables 6 and 15 report the rerun with the selected icons.

H. Reasoning Mode

Reasoning is tested on development episodes only, with each episode's own text cover story and text reports. Qwen3-VL-8B-Thinking, the reasoning checkpoint of the same model family and size as Qwen3-VL-8B-Instruct, answers with its native reasoning template; the direct answers of Qwen3-VL-8B-Instruct on the same items are the reference. Because the two are different checkpoints, the comparison bounds what reasoning adds at this scale but does not isolate reasoning within one checkpoint; for compute reasons we did not run larger reasoning models or a step-by-step prompt. The subset comprises the main task at trials 20, 40, 60 and 80 with raw and summarized history, and the described-accuracy, short and long histories and their correctness-only versions at trials 40 and 80. Generation is greedy with up to 16,384 new tokens. We report free-answer accuracy, the valid-answer rate, truncation and forced-answer rates, and token cost separately; a probability read after the generated reasoning is conditional on that reasoning and is not compared directly with the direct-answer readout. Table 17 gives the results. **[TODO: one paragraph after results]**

I. Psych-101 Protocol and Additional Fits

Task and data. In the changing bandit of Xiong et al. (2023), participants repeatedly choose between two slot machines whose payoffs stay constant until a change point, which occurs with a hazard rate h; payoffs are noise-free, so a normative learner forgets the value of the unchosen machine at rate $\phi = h$. We use the 30 participants in Psych-101 (Binz et al. 2025) and their three 100-trial games ($h = 0.1, 0.2, 0.4$; 90 games, 9,000 trials). Participants were told h, and each 100-trial game directly preceded a 1,500-trial game with the same h, so by their second and third 100-trial games participants had played 1,600 and 3,200 trials; models see each game on its own.

Replay. Models are conditioned on each participant's previously executed behavior (Binz et al. 2025): before trial t the model sees the original instructions and the participant's choices and payoffs on trials $1, \ldots, t-1$, and is asked which machine to press (one letter). The readout is the choice probability between the two letters from the first-token logits, as in the main task. In the primary condition the hazard rate is removed from the instructions (hidden); in a secondary condition the original instructions state it (told). Model and participant thus share the within-game history but not the hazard information (hidden) or the prior experience (both conditions).

Fits. We fit the myopic model-free learner of Xiong et al. (2023), which updates the chosen machine's value and forgets the unchosen one toward the midpoint at rate ϕ, with a perseveration term, separately for each hazard rate. Humans give binary choices, so for comparability we draw 20 binary choice sequences from the model's choice probabilities in each game, fit each, and take the median. The primary statistic is each participant's slope of ϕ on h; the comparison is the paired difference between the model's and the human's slope (Wilcoxon test, participant-bootstrap 95% CI), reported next to the model told versus human comparison and the model's told minus hidden contrast. The human fits are our re-analysis; Xiong et al. (2023) report no learning rates by hazard rate. In our re-analysis, the median human ϕ rose from .079 ($h = 0.1$) to .110 and .255 ($h = 0.4$; per-participant slope 0.47 per unit h, positive in 73% of participants, Wilcoxon $p = .008$), adjusting for game position did not remove the increase (0.64 [0.23, 1.05]), and in the following 1,500-trial games the increase was weaker (median ϕ .076, .114, .131; slope test $p = .029$).

Calibration of the human–model comparison. We simulated the complete pipeline 200 times per condition. Simulated participants follow each real participant's fitted strategy at each hazard rate and play on the real value sequences; the simulated model follows the same strategy with its forgetting-rate slope shifted by δ and is replayed on the simulated participant's history; both are then fitted exactly as the real data

Synthetic agent	Best strategy	κ	Repeat w	Lag-1 w
Repeat last outcome ($w = 3$)	Repeat last outcome	1.00	3.00	0.00
Delta rule ($\alpha = 0.3$, $\beta = 8$)	BOCPD	0.94	0.00	1.20
50% copy, 50% delta	Repeat last outcome	0.49	1.21	0.56
Last-5 counts	Last 5	1.00	0.02	1.08
All-trial counts	All-trial counts	1.00	0.05	0.27
Label bias ($w = 3$)	Label bias	1.00	0.00	0.00

Table 14: Strategy and kernel analyses applied to synthetic agents on the main episodes (development and held-out; no VLM). Best strategy: highest Cohen's κ; Repeat w and Lag-1 w: kernel regression weights on repeating the previous outcome and on the advisor correct one trial back. For the delta rule the fitted kernel decays geometrically with ratio 0.700 (true $1 - \alpha = 0.7$); the last-5 agent has a flat kernel up to lag 5. Strategies are correlated on this schedule, so the delta rule agrees best with BOCPD; the kernel, not the best-matching strategy, identifies it. Results on the robustness set were the same.

	Legibility controls			Main-task accuracy, correctness-only history		
Model	Image naming	Composite read	Chart read	Text reports	Image reports	Chart history
Qwen3-VL-8B	1.000	1.000	.630	.642	.549	–
InternVL3.5-8B	.988	1.000	.688	.639	.553	–
Gemma-3-27B	1.000	1.000	.604	.608	.534	–
Qwen3-VL-32B	1.000	1.000	.672	.627	.529	–
InternVL3.5-38B	1.000	**??**	.714	**??**	**??**	**??**

Table 15: Multimodal reports (legibility controls on development episodes, exploratory; selected icons, v3.1). Image naming: single-image naming over all cover stories (photos and icons). Main-task columns: held-out episodes with correctness-only history, which removes outcome evaluation; because every model names the images, the drop with image reports reflects their use in context, including advisor binding. Chart history: not yet run. Chart read below the prespecified 0.8 gate means chart-history results for that model are not interpreted.

(20 sampled choice sequences per model episode, median fit, paired Wilcoxon test and bootstrap CI). With $\delta = 0$, the test rejected at $p < .05$ in 6.0% of replications (Monte Carlo SE 1.7%) and the 95% CI covered the true difference in 93.5%, but the estimated difference was biased upward by about $+0.09$. Power was 0.49 for $\delta = -0.5$ and 0.59 for $\delta = +0.5$, where CI coverage was 0.87 and 0.96. We therefore treat the comparison as supporting and descriptive (main text).

Image and chart formats. With the hazard rate hidden, the changing bandit is also run with the last five payoffs shown as pictures of the machine's display instead of text (224 px), and with the whole history drawn as one scatter chart of payoff by trial for each machine (448 px). Each has a perception check: reading the number on one machine picture (four options; 450 items), and reading which machine was pressed, or which gave more points, at a given trial of the chart (540 items). We report each format's fits next to its perception check, since a model that cannot read the pictures or the chart cannot be expected to learn from them. **[TODO: results for image and chart formats and their perception checks]**

J. Robustness, Prompt Wording and Masking

[TODO: one paragraph after results]

Which part of the raw history blocks the correctness statement. Appending who was correct to the raw history recovered only part of the summary–raw gap, and the weight on repeating the previous outcome stayed large. To find out why, we kept the layout and length of that history and replaced parts of each past trial with a dash ("Day 3: Ken forecasts —; Sam forecasts —. Actual weather: rain. (Ken was correct; Sam was wrong.)"): the outcome label, the two reports, or both. With both replaced the history carries the same information as the correctness-only history at nearly the length of the raw history with correctness. On development episodes (Table 21), replacing the outcome label raised accuracy by 7.4 and 9.6 points (main effects; 95% CIs [5.2, 9.5] and [7.6, 11.5]) and lowered the weight on the previous outcome from .48–.50 to .12, whereas replacing the reports had no effect on its own ($+1.1$ and -0.2 points). With both replaced, accuracy was 1.5–2.1 points below the correctness-only history, so length and layout account for 14–20% of the difference between the raw history with correctness and the correctness-only history, and the visible outcome label for the rest. The outcome label thus elicits repeating the previous outcome even when who was correct is stated. Predictions for the held-out replication in five models were written before it was run. **[TODO: held-out replication]**

Prompt wording (exploratory, development episodes). We re-ran the four text history formats for Qwen3-VL-8B and InternVL3.5-8B with two alternative wordings; episodes, names, answer words, list order and readout were unchanged. Reworded sentences keep the original layout with different phrasing ("In trial 3, Sam said cat and Ken said dog; it turned out cat."; "Tally over the 3 previous trials: Ken was right 2 times and Sam was right 1 times."). A compact record uses

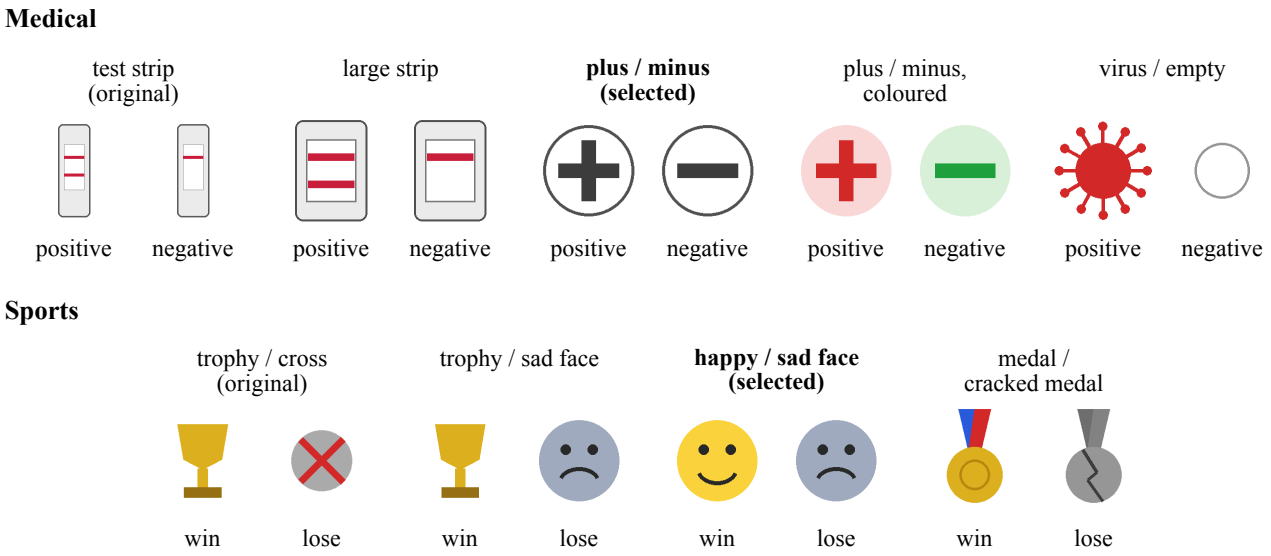


Figure 8: Actual candidate report icons for medical (positive/negative) and sports (win/lose). Original marks the v3 pairs; selected marks their v3.1 replacements. Labels and headings are shown for the reader and are not embedded in model inputs. All candidates, including those not selected, are evaluated in Table 16.

one line per trial ("#3 | Sam=cat | Ken=dog | truth=cat"; "Correct counts so far (3 trials): Ken=2, Sam=1."). Table 23 gives the results. With reworded sentences, every result was preserved: raw history at .47–.49, summary−raw +.12 and +.15, the correctness-only history at or above the summary, appending correctness recovering about 40% of the gap, and a positive weight on the previous outcome. The compact record changed the picture in both directions. Raw-history accuracy rose to .55–.56 and the weight on the previous outcome fell, whereas the correctness-only and summarized formats fell to .56–.60, and InternVL3.5-8B followed the count summary on only 72% of non-tied trials (98% with the original wording). The summary−raw gap shrank to +.03 [.01, .06] (Qwen3-VL-8B) and +.01 [−.02, .05] (InternVL3.5-8B). Raw-history accuracy stayed below the count rule (.63) in every wording, but the size of the raw–summarized gap depends on how the history is laid out, and we interpret it only for the natural-language layout used in all main conditions. The compact record is also run for the three larger models. **[TODO: larger models, compact record]**

Which part of the compact record matters. The compact record changes three parts of the prompt: the task description, the history lines together with the count summary, and the current-trial line together with the question. We crossed the original and compact version of each part (2^3 prompts; the all-original prompt reproduces the runner's prompts byte for byte; Table 22). No single part accounts for the change. Each compact part, on its own, lowered the summary−raw gap by 0.3–5.4 points (main effects of 2.7–6.3 points), and the gap stayed positive with a 95% CI above zero in all six mixed prompts. The largest single changes were these: the compact question lowered accuracy with the correctness-only history (main effect −3.6 and −4.3 points); the compact task description and count notation lowered agreement with the count rule (InternVL3.5-8B: −8.6 and −10.2 points); and for InternVL3.5-8B raw-history accuracy rose mainly when both the history lines and the current trial used the same one-line record (.477 and .491 with either alone, .559 with both). This last pattern fits the outcome-evaluation account: the gain under the raw history appeared when each past trial's reports and outcome were aligned in the same layout as the current reports, which makes deciding who was correct easier.

K. Models, Revisions and Compute

Table 24 lists the models and Table 25 the Hugging Face checkpoints and pinned revisions. A 72B model was planned as a scale extension and larger reasoning models as a supplement, but neither was run for compute reasons.

Proprietary models. GPT-6-Astra and Claude Fable 5.2 are queried through their APIs with the same prompts as the open models (text reports only), on a subset of held-out episodes and query trials, at temperature 0 where the API allows it. The choice is the answer word in the generated text; answers that are not one of the two words are counted as invalid and reported. Because the readout differs from the two-token choice probability of the open models and the subset is smaller, these results are reported separately as a reference. **[TODO: subset, number of calls, dates of the API queries, model versions]**

Image preprocessing. Icons and photos are rendered at 224 px for Qwen models and at 448 px (one tile, no tiling) for InternVL3.5. For Gemma-3, 224-px renders are resized by the processor to the fixed 896-px input (256 image tokens;

Icon pair	Naming probe	Described %	Short raw	Short corr. only	Long corr. only	Mean
Medical (positive / negative)						
Test strip (original)	.50/.50	.61/.36	.58/.62	.62/.36	.56/.45	.52
Large test strip	.50/.50	.61/.53	.58/.59	.73/.77	.62/.56	.62
Plus / minus (selected)	1.00/1.00	.70/.67	.84/.62	1.00/.50	.61/.39	.67
Plus / minus, coloured	1.00/.50	.72/.70	.77/.64	1.00/.95	.61/.52	.74
Virus / empty circle	.75/.75	.61/.92	.59/.59	1.00/.41	.48/.39	.62
Sports (win / lose)						
Trophy / cross (original)	1.00/.50	.61/.64	.52/.55	.72/.62	.61/.44	.59
Trophy / sad face	1.00/1.00	.66/.73	.55/.48	.84/.83	.61/.44	.64
Happy / sad face (selected)	1.00/1.00	.84/.78	.50/.52	.80/1.00	.50/.48	.68
Medal / cracked medal	1.00/.75	.61/.64	.53/.41	.64/.39	.48/.50	.53

Table 16: Icon comparison on development items, image reports (Qwen3-VL-8B / InternVL3.5-8B). Naming probe: accuracy over two labels $\times$ two answer orders (four prompts per model and icon pair). Downstream columns: hard accuracy over 64 items per cell after averaging counterbalanced-copy logits; Described % denotes described accuracy. Mean averages the four downstream conditions and both models, excluding the naming probe. Selection requires 4/4 naming accuracy for both models, then the highest mean.

	Free-answer accuracy							
Model / mode	Raw	Summary	Summary−raw	Described	Valid	Truncated	Forced	Median tokens
Qwen3-VL-8B-Instruct, direct	**??**	**??**	**??**	**??**	–	–	–	1
Qwen3-VL-8B-Thinking	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**??**

Table 17: Reasoning subset (development episodes; main task at trials 20, 40, 60, 80; controls at 40 and 80). Direct: the instruction-tuned checkpoint's single-token answer on the same items. Accuracy is over valid free answers; Valid, Truncated and Forced are proportions of all items.

pan-and-scan off). The correctness-only chart is rendered at 448 px for all models.

Software and hardware. All reported results on held-out episodes, the robustness set and Psych-101, the control conditions, and the development runs used to select the window length were run on NVIDIA H200 GPUs (141 GB), one GPU per run, with PyTorch 2.8.0 (CUDA 12.8) and Transformers 4.57.1. The exploratory development checks (icon selection, prompt wording, masking) were run on NVIDIA RTX 3090 GPUs with the same model revisions, precision and readout. Each run stores the rendered prompt hash, the raw candidate logits, the probability mass on the two answer tokens, and a fingerprint of the configuration. GPU-hours are in Table 26.

Code and data availability. We release the episode generator with fixed seeds, the prompts, the runners and the analysis code. Third-party data are not redistributed: a script downloads the Oxford-IIIT Pet photographs (CC BY-SA 4.0) and the Psych-101 slice of Xiong et al. (2023) (Apache-2.0), and the raw data of that study, which have no declared license, are used for analysis only. Generated episodes are checked against stored SHA-256 hashes.

Readout. Models run in bfloat16 with a float32 output projection. We read the logits of the first generated token for the two answer words, before any logit processing, and normalize them with a softmax over these two tokens to obtain y_t, the choice probability for advisor A's current report. Because the normalization ignores all other tokens, y_t is neither the model's free-generation choice probability nor a belief. Decoding is greedy, and reasoning (thinking") is switched off wherever the chat template offers it. On text-report trials with raw history we also ask a judgment question (which forecaster is currently more likely to be correct?").

Analysis plan and deviations. The analysis plan was fixed before held-out evaluation by a timestamped commit, not registered with a third party. Before that commit we had seen pilot results on an earlier version of the episodes (Appendix L), part of one model's control-condition results on development episodes, the rule-only baselines (no model) on held-out episodes, and the outputs of plumbing tests on a few held-out items, which were not analysed. After the commit, and before any held-out model output existed, the hardware used for evaluation changed and a 72B model and a 32B reasoning model were dropped, both for compute reasons; the selected icons for two cover stories were replaced after development checks (Appendix G). The masking experiment (Appendix J), the prompt-wording checks and the Bayesian reference learner were added after the commit and are exploratory.

Readout quality. Because y_t is normalized over the two candidate tokens only, we report for every model, format and condition the probability mass on the two candidates (mean, minimum and share of trials below 0.5) and the format-following rate (share of trials on which the greedy output is one of the two answer words); see Table 26.

		Median forgetting rate ϕ				Slope difference (model − human)	
Agent	Hazard info	$h = 0.1$	$h = 0.2$	$h = 0.4$	Slope	Est. [95% CI]	p
Humans ($n = 30$)	told	.079	.110	.255	0.47	–	–
Qwen3-VL-8B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
InternVL3.5-8B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
Gemma-3-27B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
Qwen3-VL-32B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??
InternVL3.5-38B	hidden	??	??	??	??	??	??
	told	??	??	??	??	??	??

Table 18: Psych-101 changing bandit (Xiong et al. 2023). Models are conditioned on each participant's history; ϕ is fitted to choices sampled from the model (20 draws, median). Slope: median over participants of the per-participant slope of ϕ on hazard rate (humans: our re-analysis). The paired test has power of about 0.5 and an upward bias of about $+0.09$ (calibration above), so it is descriptive only.

Model	Raw	Summary	Gap [95% CI]	p
Qwen3-VL-8B	??	??	??	??
InternVL3.5-8B	??	??	??	??
Gemma-3-27B	??	??	??	??
Qwen3-VL-32B	??	??	??	??
InternVL3.5-38B	??	??	??	??

Table 19: Primary comparison on the robustness set (24 episodes with independent outcomes and irregular reversals).

L. Pilot Results

Before the analysis plan was fixed, we ran a pilot on RTX 3090 GPUs with an earlier version of the episodes (16 episodes), whose instructions always introduced the advisors in the same order; this confound was removed in the episodes used here, so the pilot only motivated the design. In Qwen3-VL-32B, the change-point summary exceeded the raw history by $+.191$ [.161, .220] and the all-trial summary by $+.145$; the model given a summary matched the rule alone ($+.001$); under raw histories its choices agreed most with repeating the previous outcome ($\kappa = 0.25$) and its integration kernel was nearly flat. With photo reports it followed the advisor listed first (1.00 when advisor A's photo came first, .03 when second), and Qwen3-VL-8B read single cells of the correctness chart on .56 of items. These observations shaped the control conditions, the binding checks and the chart gate; none of the pilot data enter the results.

	Text cover stories							
Model	Color	Weather	Stock	Sports	Medical	News	Sign test p	Pet (photo)
Qwen3-VL-8B	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??

Table 20: Summary−raw gap by cover story (held-out; eight episodes per text cover story). Pet uses photo reports in its own episode slot and is not part of the sign test.

	Raw + correctness		Reports shown		Outcome shown		Both replaced		Correctness only	
Model	Acc.	Repeat	Acc.	Repeat	Acc.	Repeat	Acc.	Repeat	Acc.	Repeat
Development episodes										
Qwen3-VL-8B	.534	.50	.576	.12	.513	.48	.620	.04	.640	.03
InternVL3.5-8B	.530	.48	.602	.12	.504	.27	.625	.02	.640	.01
Held-out episodes										
Qwen3-VL-8B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-8B	??	??	??	??	??	??	??	??	??	??
Gemma-3-27B	??	??	??	??	??	??	??	??	??	??
Qwen3-VL-32B	??	??	??	??	??	??	??	??	??	??
InternVL3.5-38B	??	??	??	??	??	??	??	??	??	??

Table 21: Masking parts of the raw history with correctness (text reports; accuracy averaged over counterbalanced copies, $t \geq 2$). Reports shown: outcome label replaced; Outcome shown: reports replaced; Both replaced: same information as the correctness-only history. Repeat: regression weight on repeating the previous outcome. Development rows are exploratory (RTX 3090); held-out predictions were fixed beforehand: replacing the outcome label raises accuracy, replacing both beats the raw history with correctness, and replacing the outcome label matters more than replacing the reports.

			Qwen3-VL-8B				InternVL3.5-8B			
Task	History	Question	Raw	Corr. only	Summ.	Summ.−raw [95% CI]	Raw	Corr. only	Summ.	Summ.−raw [95% CI]
O	O	O	.485	.640	.617	+.132 [.108, .153]	.471	.640	.628	+.157 [.140, .174]
O	O	T	.512	.613	.627	+.115 [.093, .137]	.491	.604	.617	+.126 [.106, .146]
O	T	O	.494	.633	.623	+.129 [.114, .145]	.477	.637	.600	+.122 [.099, .145]
O	T	T	.532	.588	.608	+.076 [.051, .105]	.559	.616	.597	+.038 [.007, .065]
T	O	O	.506	.637	.600	+.094 [.066, .121]	.507	.634	.610	+.103 [.073, .132]
T	O	T	.520	.599	.600	+.080 [.052, .108]	.493	.562	.600	+.107 [.079, .131]
T	T	O	.526	.616	.600	+.074 [.040, .104]	.515	.634	.583	+.068 [.035, .100]
T	T	T	.563	.580	.597	+.034 [.008, .059]	.550	.593	.564	+.014 [−.019, .045]

Table 22: Decomposition of the compact record (development episodes, text reports; accuracy averaged over counterbalanced copies, $t \geq 2$). O: original wording of that part; T: compact version. Task: task description; History: history lines and count summary; Question: current-trial line and question. The first row is the original prompt and the last row the compact record of Table 23.

Model	Wording	Raw	Raw+c.	Corr. only	Summ.	Summ.−raw [95% CI]	Repeat
Qwen3-VL-8B	Original	.485	.534	.640	.617	+.132 [.108, .153]	.46
	Reworded	.488	.533	.624	.607	+.119 [.093, .144]	.50
	Compact	.563	.561	.580	.597	+.034 [.008, .059]	.32
InternVL3.5-8B	Original	.471	.530	.640	.628	+.157 [.140, .174]	.28
	Reworded	.473	.538	.644	.627	+.153 [.133, .175]	.26
	Compact	.550	.559	.593	.564	+.014 [−.019, .045]	.13

Table 23: Prompt-wording check (development episodes, text reports; accuracy averaged over counterbalanced copies, $t \geq 2$). Repeat: regression weight on the previous outcome under the raw history.

Model	LM family	Params	Image input
Qwen3-VL-8B	Qwen3	8B	224 px
InternVL3.5-8B	Qwen3	8B	448 px
Gemma-3-27B	Gemma 3	27B	896 px[a]
Qwen3-VL-32B	Qwen3	32B	224 px
InternVL3.5-38B	Qwen3	38B	448 px
Qwen3-VL-8B-Thinking[b]	Qwen3	8B	224 px

Table 24: Models (used without fine-tuning; pinned revisions in Table 25). Top block: core models (instruction-tuned). InternVL3.5: one 448-px tile. [a]224-px renders resized by the processor. [b]Reasoning subset only.

Model	Checkpoint	Revision
Qwen3-VL-8B	Qwen/Qwen3-VL-8B-Instruct	0c351dd01ed87e9c1b53cbc748cba10e6187ff3b
InternVL3.5-8B	OpenGVLab/InternVL3_5-8B-HF	741a7d03020411e666c6109218ab71e08151ef86
Gemma-3-27B	google/gemma-3-27b-it	005ad3404e59d6023443cb575daa05336842228a
Qwen3-VL-32B	Qwen/Qwen3-VL-32B-Instruct	0cfaf48183f594c314753d30a4c4974bc75f3ccb
InternVL3.5-38B	OpenGVLab/InternVL3_5-38B-HF	7c830fc25e874f9a861087e3c70587f2b9b555f3
Qwen3-VL-8B-Thinking	Qwen/Qwen3-VL-8B-Thinking	92f3c4b4feadd3a016ef468d103bb5f58b2a2c6b

Table 25: Checkpoints and pinned revisions (resolved 2026-11-11).

	Candidate probability mass			Format	
Model	Mean	Min	Share < 0.5	following	GPU-hours
Qwen3-VL-8B	1.000	.997	.000	1.000	**??**
InternVL3.5-8B	.996	.112	.002	.996	**??**
Gemma-3-27B	**??**	**??**	**??**	**??**	**??**
Qwen3-VL-32B	.996	.907	.000	1.000	**??**
InternVL3.5-38B	**??**	**??**	**??**	**??**	**??**
Qwen3-VL-8B-Thinking	**??**	**??**	**??**	**??**	**??**
Total					**??**

Table 26: Readout quality pooled over formats and conditions per model (per cell in the released analysis output). **[DRAFT: development episodes (16) and four models; replace with held-out results]** (main task, four history formats).